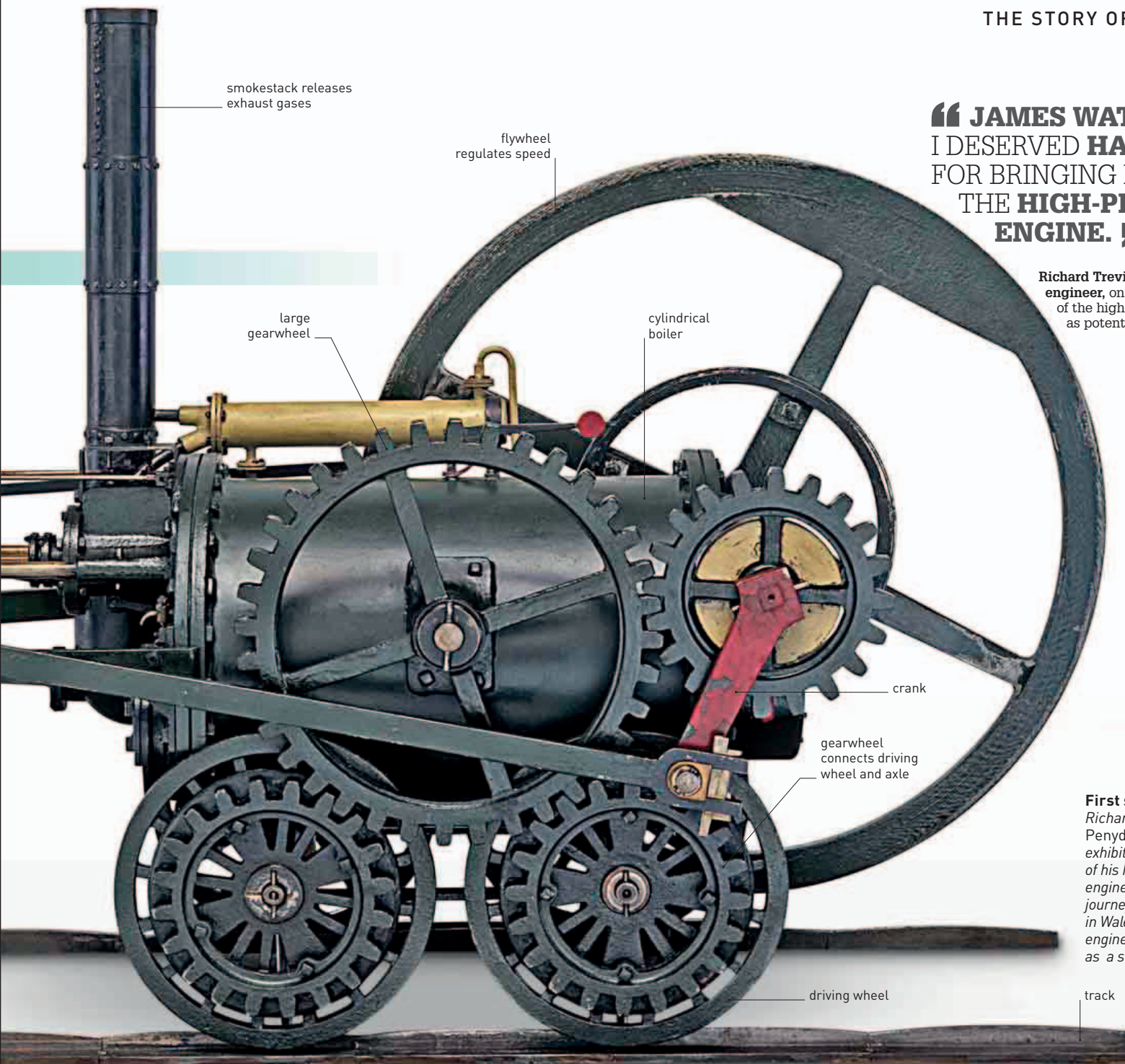


**“ JAMES WATT [SAID]...
I DESERVED **HANGING**
FOR BRINGING INTO USE
THE **HIGH-PRESSURE**
ENGINE. ”**

Richard Trevithick, English engineer, on Watt's perception of the high-pressure engine as potentially explosive, 1833



smokestack releases exhaust gases

flywheel regulates speed

large gearwheel

cylindrical boiler

crank

gearwheel connects driving wheel and axle

driving wheel

track

First steam locomotive

Richard Trevithick's Penydarren locomotive exhibited the effectiveness of his high-pressure steam engine when it made its first journey on February 21, 1804 in Wales. The locomotive's engine could also be used as a stationary engine.

1860

Gas engine

Invented by Belgian engineer Étienne Lenoir, the first successful internal combustion engine—the gas engine—generates power by burning gas and air inside a cylinder.



Étienne Lenoir's gas engine

1897

Diesel engine

The first diesel engine is built by French-German engineer Rudolf Diesel. Despite being heavier, his invention is more efficient than gasoline engines, and uses the heat of compression, rather than a spark, to ignite fuel.



Diesel engine

1937

Turbojet engine

English engineer Frank Whittle and German engineer Hans von Ohain independently develop and test engines that burn fuel and use a fan to fire out a continuous jet of hot air to thrust a plane forward.



W2/700 Turbojet engine

1816

Closed-cycle steam engine

Scottish engineer Robert Stirling invents a steam engine in which gases remain within the system, so there is no exhaust and little noise from explosions.

1876

Four-stroke engine

German engineer Nikolaus Otto's powerful, four-stroke cycle engine fires four cylinders in turn, so fuel ignited in each pushes down the piston on every fourth stroke.



Four-stroke Daimler motorcycle

1926

Liquid fuel rocket

American engineer Robert Goddard's invents the rocket engine. Thrust is achieved for flight by burning liquid fuel.

1956

Rotary engine

German engineer Felix Wankel creates a rotary engine that has a triangular rotor inside an oblong cylinder instead of pistons.



Wankel's rotary engine